

13	The density of argon gas at a pressure of $1.00 \times 10^5 \text{ Pa}$ is 1.60 kg m^{-3}. What is the root-mean-square speed of the argon molecules?							
	A	216 m s^{-1}	B	250 m s^{-1}	C	306 m s^{-1}	D	433 m s^{-1}

Answer: D

$pV = \frac{1}{3}Nm\langle c^2 \rangle$ where p is the pressure, V the volume, N the number of molecules, m the mass and $\langle c^2 \rangle$ the mean square speed of molecules.

Therefore, $p = \frac{1}{3}\rho\langle c^2 \rangle$ where ρ is the density of molecules.

$$\langle c^2 \rangle = \frac{3p}{\rho}$$

$$\sqrt{\langle c^2 \rangle} = \sqrt{\frac{3p}{\rho}} = \sqrt{\frac{3(1.00 \times 10^5)}{1.60}} = 433 \text{ m s}^{-1}$$

